



SM

SHEHAB
M. ABDELSALAM

ECE Student || Service Provider Networks ||

Network Security || Cloud Infrastructure ||

Network Automation

 [linkedin.com/in/shehab-m-abdelsalam](https://www.linkedin.com/in/shehab-m-abdelsalam)

 shehababdelsalam1@gmail.com

VRRP

[*Virtual Router Redundancy Protocol*]

GNS3 || Cisco IOS

DOCUMENT

Technical Documentation

VERSION

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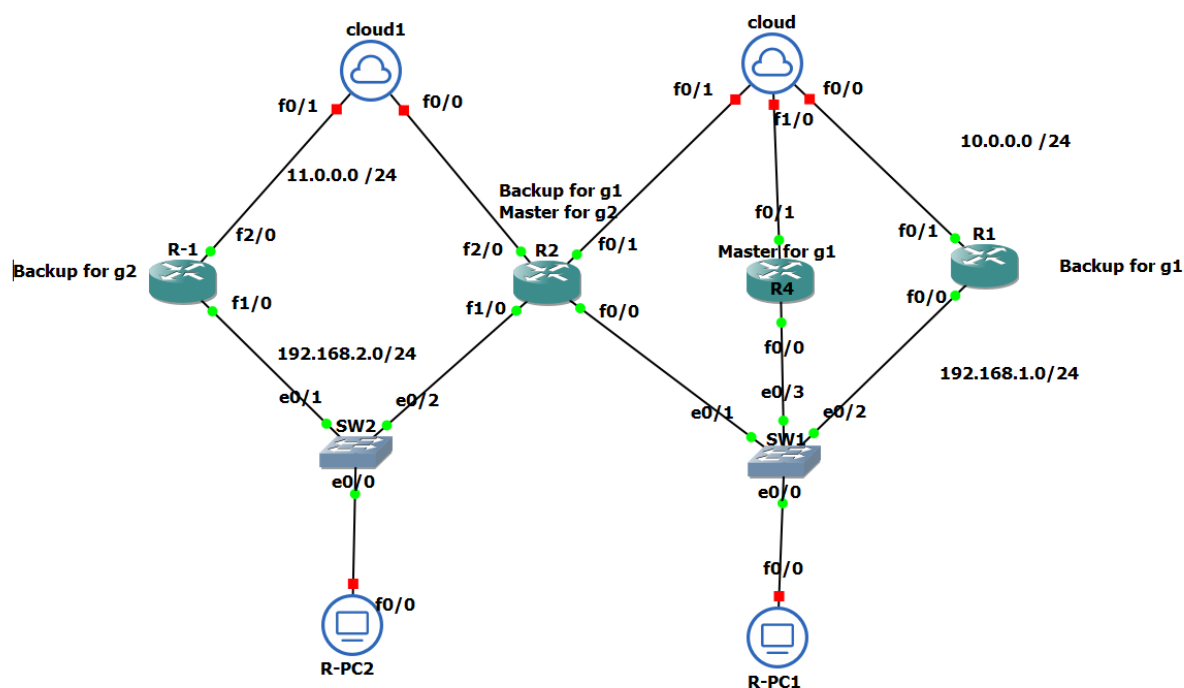
Virtual Router Redundancy Protocol (VRRP) Lab Manual

Dual-Group Implementation, MD5 Authentication & Object Tracking Analysis

1. Project Overview

This laboratory document details the implementation and configuration of the Virtual Router Redundancy Protocol (VRRP), an open-standard First-Hop Redundancy Protocol (FHRP). The project demonstrates high availability, MD5 authentication security, and object tracking mechanisms across a multi-router topology to provide seamless failover and active/active load distribution between distinct subnets.

2. Network Topology



3. Addressing Scheme

Device	Interface	IP Address / Subnet	Role / Description
R1	FastEthernet0/0	192.168.1.1 /24	Group 1 Backup Router (Priority 105) WAN 1 Link Tracked Loopback
	FastEthernet0/1	10.0.0.1 /24	
	Loopback0	1.1.1.1 /32	
R4	FastEthernet0/0	192.168.1.4 /24	Group 1 Master Router (Priority 150) WAN 1 Link
	FastEthernet0/1	10.0.0.4 /24	
R2	FastEthernet0/0	192.168.1.2 /24	Group 1 Backup Router (Priority 100) Group 2 Master Router (Priority 150) WAN 2 Link
	FastEthernet1/0	192.168.2.2 /24	
	FastEthernet2/0	11.0.0.2 /24	
R3	FastEthernet1/0	192.168.2.1 /24	Group 2 Backup Router (Priority 100) WAN 2 Link
	FastEthernet2/0	11.0.0.3 /24	
R-PC1	FastEthernet0/0	192.168.1.200 /24	Host in LAN 1 (Virtual Gateway: 192.168.1.3)
	FastEthernet0/0	192.168.2.200 /24	Host in LAN 2 (Virtual Gateway: 192.168.2.3)

4. Technical Specifications & VRRP Mechanics

VRRP Group 1 Specification (LAN 1 - 192.168.1.0/24)

- Virtual IP Address: 192.168.1.3
- Authentication: MD5 Key-String (123)
- Primary Master Router: R4 (Configured Priority: 150)
- Backup Routers: R1 (Priority: 105), R2 (Default Priority: 100)
- Interface Tracking & Failover Logic: R4 tracks FastEthernet0/1 (Track 300). Upon failure of FastEthernet0/1, R4 decrements its priority by 60 points (150 -> 90). Because VRRP has preemption enabled by default, R1 (Priority 105) automatically assumes the Master Router role. R1 also tracks FastEthernet0/1 (Track 300) with a decrement value of 15

VRRP Group 2 Specification (LAN 2 - 192.168.2.0/24)

- Virtual IP Address: 192.168.2.3
- Authentication: MD5 Key-String (123)
- Primary Master Router: R2 (Configured Priority: 150)
- Primary Backup Router: R3 (Default Priority: 100)
- Interface Tracking & Failover Logic: R2 tracks FastEthernet2/0 (Track 500). When FastEthernet2/0 goes down, R2's priority decrements by 60 points (150 -> 90). R3 (Priority 100) then preempts R2 and takes over active forwarding for VRRP Group 2.

5. Device Configurations

Router R1 Configuration

```
hostname R1
!
track 200 interface Loopback0 line-protocol
track 300 interface FastEthernet0/1 line-protocol
!
interface Loopback0
 ip address 1.1.1.1 255.255.255.255
!
interface FastEthernet0/0
 ip address 192.168.1.1 255.255.255.0
 vrrp 1 ip 192.168.1.3
 vrrp 1 priority 105
 vrrp 1 authentication md5 key-string 123
 vrrp 1 track 300 decrement 15
!
interface FastEthernet0/1
 ip address 10.0.0.1 255.255.255.0
!
```

Router R4 Configuration

```
hostname R4
!
track 300 interface FastEthernet0/1 line-protocol
!
interface FastEthernet0/0
 ip address 192.168.1.4 255.255.255.0
 vrrp 1 ip 192.168.1.3
 vrrp 1 priority 150
 vrrp 1 authentication md5 key-string 123
 vrrp 1 track 300 decrement 60
!
interface FastEthernet0/1
 ip address 10.0.0.4 255.255.255.0
!
```

Router R2 Configuration

```
hostname R2
!
track 500 interface FastEthernet2/0 line-protocol
!
interface FastEthernet0/0
 ip address 192.168.1.2 255.255.255.0
 vrrp 1 ip 192.168.1.3
 vrrp 1 authentication md5 key-string 123
!
interface FastEthernet1/0
 ip address 192.168.2.2 255.255.255.0
 vrrp 2 ip 192.168.2.3
 vrrp 2 priority 150
 vrrp 2 authentication md5 key-string 123
 vrrp 2 track 500 decrement 60
!
interface FastEthernet2/0
 ip address 11.0.0.2 255.255.255.0
!
```

Router R3 Configuration

```
hostname R3
!
interface FastEthernet1/0
 ip address 192.168.2.1 255.255.255.0
 vrrp 2 ip 192.168.2.3
 vrrp 2 authentication md5 key-string 123
!
interface FastEthernet2/0
 ip address 11.0.0.3 255.255.255.0
!
```

6. Verification and Diagnostic Commands

Use the following Cisco IOS commands to verify VRRP state transitions, tracking, and authentication status:

```
show vrrp
show vrrp brief
show track
```